

HONORS PHYSICS: KINEMATIC
RENDEZVOUS CHALLENGE

HS-PS2-1 • SEP-5

Names: Period: Date:
0

UNIT 2: 1D CONSTANT VELOCITY & DELAYED INTERCEPT
MODELING

MISSION DIRECTIVE: HARBOR EMERGENCY MEDICAL AIR-DROP

A marine research vessel has departed the harbor dock (designated as reference origin x = 0 m) and is steaming straight out into open waters. At mission start (t = 0 s), the vessel is already 300 meters out from the dock and cruising steadily at v_ship = 6.0 m/s. An urgent medical distress call requires dispatching an autonomous high-speed courier drone from the harbor dock (x_drone = 0 m). The drone cruises at a uniform speed of v_drone = 18.0 m/s, but requires a 20-second flight-preflight delay (system reboot and rotor spin-up) before it launches from the dock at t = 20 s.

Marine Research Vessel Telemetry

- Reference Origin: Dock (x = 0 m)
Initial Position at t = 0 s:
x_o_ship = +300 m
Uniform Cruise Speed:
v_ship = +6.0 m/s
Launch Delay: None (in motion at t = 0 s)

Autonomous Medical Drone Telemetry

- Reference Origin: Dock (x = 0 m)
Launch Position at t = 20 s:
x_o_drone = 0 m
Uniform Cruise Speed:
v_drone = +18.0 m/s
Preflight Delay:
t_delay = 20.0 seconds

PART 1: ALGEBRAIC KINEMATIC MODELING & PREDICTION

TEAM ALGEBRAIC PROOF

Step 1A: Formulate the continuous Position-Time equation for the Research Vessel:

Write x_ship(t) in terms of t: x_ship(t) =

Step 1B: Formulate the piecewise Position-Time equation for the Drone (for t ≥ 20 s):

Factored Form: x_drone(t) = 18.0 · (t - 20) → Expanded Form: x_drone(t) =

Step 1C: Equate positions (x_ship = x_drone) to solve for exact Rendezvous Time (t_meet) and Position (x_meet):

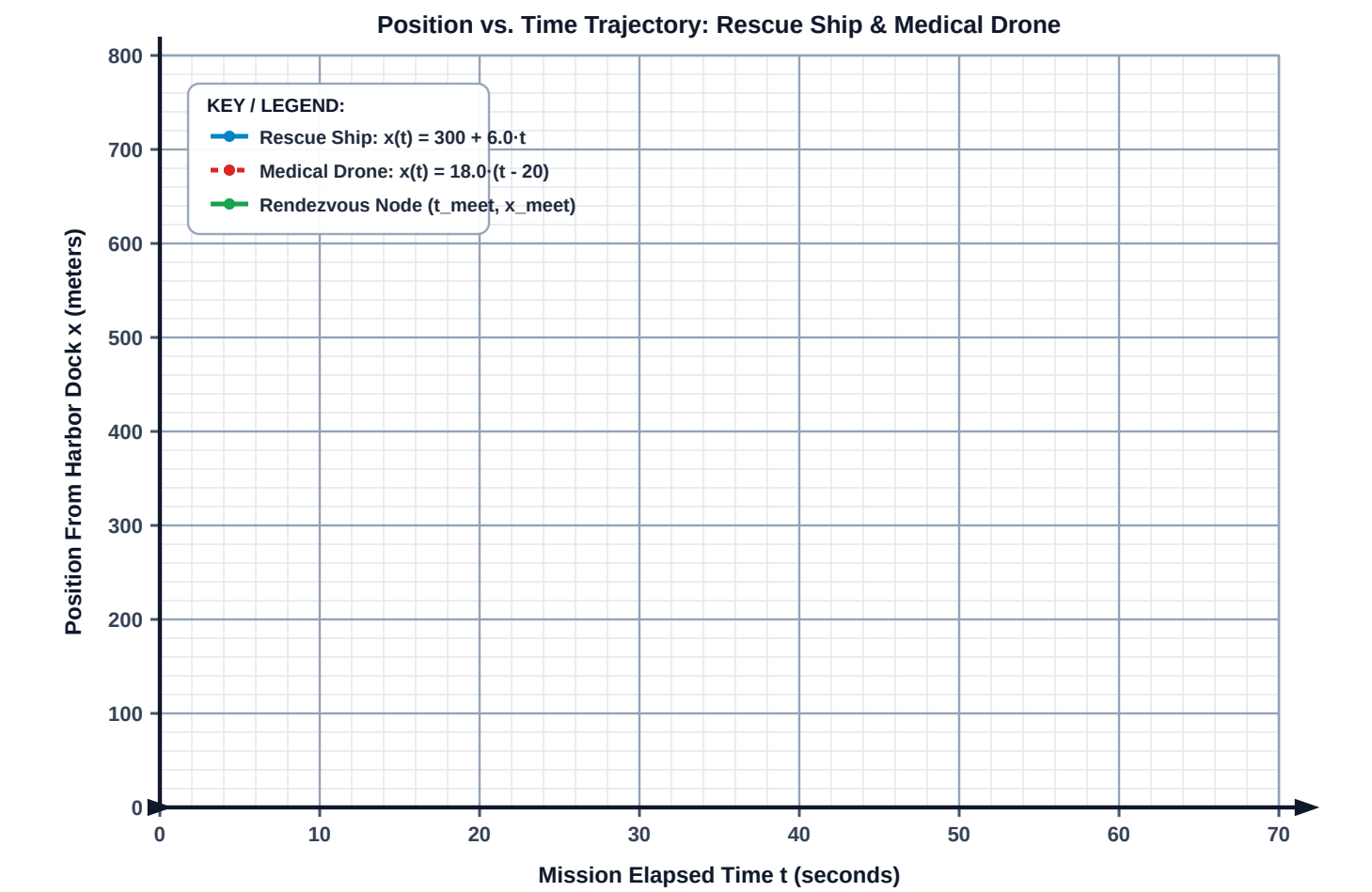
Show step-by-step algebraic isolation of time t:

Rendezvous Time: t = seconds Rendezvous Position: x = meters from dock

PART 2: MULTI-AGENT COORDINATE TELEMETRY TABLE

PLOTTING DATA POINTS

Table with 9 columns: Vehicle, t = 0 s, t = 10 s, t = 20 s (Launch), t = 30 s, t = 40 s, t = 50 s, t = 55 s (Meet), t = 60 s. Rows include Ship Position x (m) and Drone Position x (m).



PART 4: MISSION ENGINEERING AUDIT & CONSTRAINT ANALYSIS

HONORS SYNTHESIS

1. Slope Comparison & Visual Verification:

Calculate the slope of each graphed line using $(\Delta x / \Delta t)$. Does the intersection point on your graph exactly match your algebraic calculation from Page 1? Identify any differences:

2. Flight Duration vs. Mission Elapsed Time:

The rendezvous occurs at mission clock $t = 55$ seconds. How many seconds was the drone actually flying in the air? Show the subtraction and explain why elapsed mission time differs from flight duration:

3. Battery Range Safety Audit (The Critical Failure Test):

The drone's lithium-polymer battery pack certifies a maximum round-trip total flight distance of **1,200 meters** before complete depletion. Can the drone deliver the package to the ship at the rendezvous point and safely return to the dock? **Prove your claim with numbers:**